

Assessify: Comprehensive UI, UX & Business Process Analysis

Internal Assessment System & Evaluation Engine · Yayasan Karya Bangsa

Date: September 2026 · Standard: CEFR English Placement (A1–C1) · Version: 2.5 Live

1. Executive Summary

Assessify is a custom-engineered, full-stack English placement assessment and evaluation management platform built for Yayasan Karya Bangsa. The system standardizes English diagnostic testing, automated answer-key evaluation, and administrative qualitative scoring across all five foundation school units: KB-TK Golden Bee, SD Karya Bangsa, SMP Karya Bangsa, SMA Karya Bangsa, and SMK Karya Bangsa. This report delivers an exhaustive audit of the platform's visual user interface (UI), user experience (UX) workflows, and underlying enterprise business processes based on live testing.

2. User Interface (UI) Design Analysis

2.1. Color System & CEFR Visual Standards

The user interface applies a refined corporate-academic visual hierarchy. Primary actions utilize Royal Blue (#2563eb) and Deep Navy (#0f274a) for authority and focus, while candidate placement bands strictly follow standardized, high-contrast semantic palettes:

C1 Advanced (18–20 pts)	Royal Purple (#7c3aed / #f5f3ff) · Highest tier mastery indicator
B2 Upper-Intermediate	Emerald Green (#059669 / #ecfdf5) · Strong independent command
B1 Intermediate (10–13 pts)	Royal Blue (#2563eb / #eff6ff) · Operational foundation proficiency
A2 Elementary (7–9 pts)	Warm Amber (#d97706 / #fffbeb) · Developing core language structures
A1 Beginner (4–6 pts)	Crimson Red (#c0392b / #fef2f2) · Foundational learning threshold

2.2. Component Architecture & Micro-Interactions

- Pre-test Hardware Status Bar: Features real-time SVG status indicators with dynamic pulse animations (green for active streams, blue during diagnostic cycles, and red upon device denial).
- Retest Device Badge Button: Upgraded from plain underlined text to an interactive pill badge (border-radius: 20px, light blue background, hover elevation, and spinning SVG loader during active stream re-acquisition).
- Dynamic Real-time Letter Counter: Integrated live character counter badge inside writing textareas with instant visual validation (transforms into a green checkmark pill when meeting recommended target lengths).
- Floating Bulk Action Toolbar: High-contrast navy toolbar (#0f274a) that smoothly animates into view upon checkbox selection in the Admin Results table.

3. User Experience (UX) Architecture Analysis

3.1. Educator & Candidate Assessment Journey

The candidate experience was designed to eliminate cognitive friction, test anxiety, and technical impediments before and during the timed assessment:

Portal Selection & State Retention	Login workspace dropdown cleanly separates Teacher Assessment from School Admin. Preference is cached in localStorage to prevent reset on page refresh.
Pre-Test Readiness & Diagnostics	Live camera visualizer, Audio Volume, and 5-second sample recording verify media devices before the timed test timer begins, preventing lost test minutes.
Timed 75-Minute Global Clock	Fixed topbar timer tracks real-time progress across all 5 sections. Section steppers clearly indicate completed (✓) and active phases.
Single-Play Audio Prompts	Elstering audio prompts enforce single-play security with visual status locking and yellow "Playing Audio..." loading feedback.
Speaking Capture Workflow	Direct browser media recording captures audio and video without requiring manual transcription, saving candidate effort and preventing typing errors.

3.2. Administrative Review & Grading Experience

The Admin Dashboard consolidates complex diagnostic telemetry into actionable, high-velocity workflows:

3-Column Executive KPI Cards	Provides instant visibility of Total Attempts, Completed Tests, and Pending Reviews at a glance.
Multi-Unit Filter & Search	Dropdown allows single-click filtering between All Units and the 5 specific school units, instantly synchronizing URL parameters for Excel and PDF exports.
Interactive Batch Table	Includes header "Select All" checkbox with indeterminate states, individual row checkboxes, and rapid batch actions (Export Selected Excel, Export PDF, Bulk Delete).
Full Rubric Evaluation Modal	Streamlined dual-pane grading modal presenting the candidate's written essay and video recording alongside interactive 1–5 segmented scale buttons for instant composite scoring.

4. Business Process & Enterprise Workflow Analysis

4.1. Foundation School Unit Governance Matrix

Assessify unifies language proficiency benchmarking across the entire foundation while maintaining strict unit data isolation and tailored reporting:

KB-TK Golden Bee	Early Childhood & Kindergarten Division · Foundational language metrics
SD Karya Bangsa	Primary School Division · Elementary curriculum placement & literacy development
SMP Karya Bangsa	Junior High School Division · Intermediate communicative competency
SMA Karya Bangsa	Senior High School Division · Upper-intermediate academic English proficiency
SMK Karya Bangsa	Vocational High School Division · Practical and professional industry readiness

4.2. Dual-Track Hybrid Evaluation Workflow

The evaluation engine combines automated objective answer-key verification with human administrative review to synthesize certified CEFR placements:

Track A: Objective Auto-Scoring	Grammar (20 items), Reading (15 items), and Listening (15 items) are graded synchronously upon submission against verified answer keys.
Track B: Human Qualitative	Writing (2 tasks) and Speaking (2 interview prompts) are routed to administrator review, evaluated against 4 standard criteria on a 1–5 scale (Max 20 pts each).
Composite Synthesis & Placement	The system calculates the overall CEFR placement (A1–C1) using public thresholds and automated consistency checks once all 5 skills are graded.
Export & Official Certification	Produces instant official PDF placement certificates with school watermarks and formatted Excel rosters for Yayasan board evaluation.

5. Live Verification & Quality Assurance Summary

All core system flows were executed and verified on the live system on September 1, 2026. The test suite validated 9 primary modules with 100% pass rate:

Module 1: Server & Database Health	PASSED · Node.js runtime connected to MySQL InnoDB repository
Module 2: 75-Min Assessment Structure	PASSED · 15 Listening, 20 Grammar, 15 Reading, 2 Writing, 2 Speaking (54 items)
Module 3: Multi-Unit Authentication	PASSED · Unit selection enforced, session state preserved across reloads
Module 4: Candidate Attempt Lifecycle	PASSED · Attempt creation, timed submission, and media upload pipeline
Module 5: Admin Evaluation & C1 Scale	PASSED · 4-criteria rubric grading evaluated up to C1 level (18–20 pts)
Module 6: Single & Bulk Exports	PASSED · Dynamic Excel (.xlsx) spreadsheet and centered PDF reports generated

